

Linear systems and elimination

A sharp subset bound for worst-case normal GMRES

Difficulty: challenging **Importance:** interesting to the community

Status: explicit conjecture; no later resolution located

Let $L \subset \mathbb{C} \setminus \{0\}$ consist of $n \geq 3$ distinct points. For a nonempty $S \subseteq L$, define

$$M_k(S) = \min_{p \in \mathbb{C}[z], \deg p \leq k, p(0)=1} \max_{z \in S} |p(z)|.$$

Is it true, simultaneously for every such L and $1 \leq k \leq n-2$, that

$$M_k(L) \leq \frac{4}{\pi} \max_{S \subseteq L, |S|=k+1} M_k(S)?$$

The constant is required to be independent of dimension, degree, and the locations of the points.

For a normal matrix with spectrum L , $M_k(L)$ is the largest relative residual norm attainable at GMRES step k , over unit initial residuals. The smaller-set quantities have explicit formulas through Lagrange interpolation. The question therefore asks for a sharp, dimension-independent certificate of worst-case convergence using small subsets of the spectrum. It is distinct from equality of ideal and worst-case GMRES for a nonnormal Jordan block (IE-02).

References

J. Liesen and P. Tichý, *The worst-case GMRES for normal matrices*, BIT 44 (2004), 79–98, conjecture (3.16), pp. 91–92, and Appendix (author copy). Their *A min-max problem on roots of unity*, TU Berlin Preprint 28-2003, conclusion (9.1), proves selected roots-of-unity cases and sharpness of $4/\pi$ (university record). Their survey *Convergence analysis of Krylov subspace methods*, GAMM-Mitteilungen 27 (2004), discussion after (11), restates the conjecture (author copy).

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Searched the original title with “conjecture”, “ $4/\pi$ ”, “ $4/\pi$ ”, “proof”, “counterexample”, and 2025–2026; checked the authors’ current publication lists. The later Jordan-block papers concern a different conjecture. No resolution or recent explicit reaffirmation of this particular bound was located; its open-status evidence is historical and bounded.